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$$\begin{aligned} & \int_{-\infty}^{+\infty} \frac{x}{x^2+4} dx \\ &= \int_{-\infty}^0 \frac{x}{x^2+4} dx + \int_0^{+\infty} \frac{x}{x^2+4} dx \\ & u = x^2 + 4 \Rightarrow du = 2x dx \Rightarrow \frac{du}{2} = x dx \\ & \int \frac{x}{x^2+4} dx = \frac{1}{2} \ln(x^2+4) + C \\ & \int_0^{+\infty} \frac{x}{x^2+4} dx = \lim_{b \rightarrow +\infty} \left[\frac{1}{2} \ln(x^2+4) \right]_0^b \\ &= \lim_{b \rightarrow +\infty} \left(\frac{1}{2} \ln(b^2+4) - \frac{1}{2} \ln 4 \right) = +\infty \\ & \int_{-\infty}^0 \frac{x}{x^2+4} dx = \lim_{a \rightarrow -\infty} \left[\frac{1}{2} \ln(x^2+4) \right]_a^0 \\ &= \lim_{a \rightarrow -\infty} \left(\frac{1}{2} \ln 4 - \frac{1}{2} \ln(a^2+4) \right) = -\infty \\ & \int_{-\infty}^{+\infty} \frac{x}{x^2+4} dx \text{ bestaat niet} \end{aligned}$$

References